

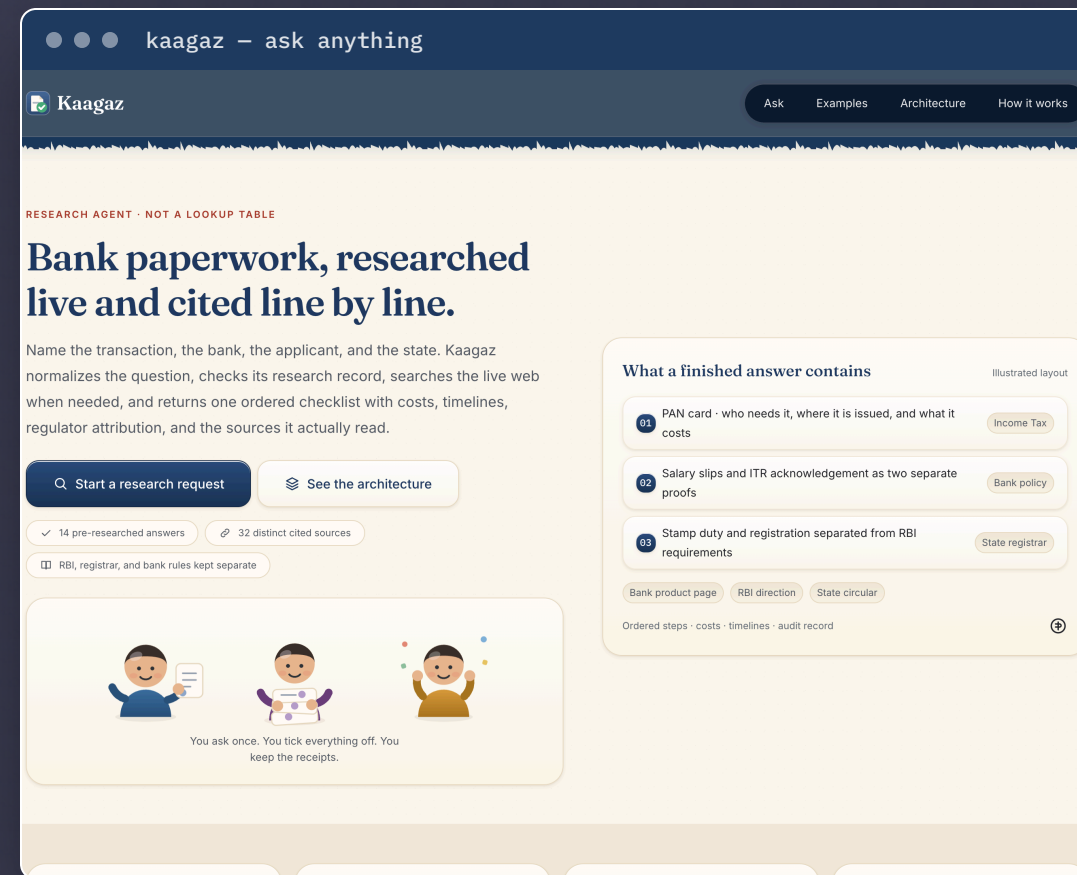
# Bank paperwork, researched live and cited line by line.

Kaagaz turns any Indian banking-document question into an ordered, costed, regulator-aware checklist — with the sources it actually read and the exact research record behind every figure.

14 pre-researched answers

32 cited sources

188 tests · 4 browser gates



# Paperwork fails at the counter.

## 🕒 Nobody finds out early

The missing affidavit, the attestation not done abroad, the stamp denomination nobody mentioned — all of it surfaces during the appointment.

## 📁 Scattered across four sources

Bank product pages, RBI circulars, state stamp-duty schedules, registrar notices. None of them agrees on a single list.

## ⚠️ The regulators get conflated

“The RBI says you need stamp duty” sends someone to entirely the wrong government office — and it is the most common error in the domain.

## 📅 No cost, no timeline

A list of names is not a plan. What it costs, how many days each step takes, and which one blocks the next is what makes it usable.

The same question has four different answers. One per residency status — and more again per state, because stamp duty and registration fees are state law.

Ask an NRI for a resident’s checklist and you get a list full of documents they do not need — and no work permit, no attested contract, no overseas statements.

## WHAT IT COSTS IN PRACTICE

Repeat visits, missed appointment slots, and occasionally a missed disbursement deadline. The information existed the whole time — it was just never in one ordered place.

# A checklist app has a ceiling set by its curator.

## THE OBVIOUS APPROACH

- ✓ Someone decides which transactions matter and hand-writes the list.
- ✓ It covers exactly those cases, and nothing outside them.
- ✓ It goes stale the week a bank revises a fee schedule.
- ✓ Nobody can tell which regulator imposed which line.

## WHAT KAAGAZ DOES INSTEAD

- ✓ Free-text question. **Unknown banks and products are researched, not rejected.**
- ✓ Search the live web, read the pages, **cite only what was opened.**
- ✓ Freshness is per entry, with an **as-of date on screen.**
- ✓ Regulator attribution is **enforced in code**, not requested in a prompt.

## QUESTIONS A FIXED LIST CANNOT ANSWER

**"Kisan Credit Card renewal for a Kisan with 2 land records"**

Not one of the four canonical products, and the document set depends on how many land records exist.

**"Gold loan at a small finance bank, not HDFC"**

A bank nobody curated. It is still researched, ranked by its own name match, and cited from its own pages.

**"Property registration in Kerala for an NRI buyer"**

State fees *and* residency rules *and* an attestation chain, all at once — the case a single-list product gets wrong most often.

**The cache is a floor, not a ceiling.** 14 answers are pre-warmed so the common cases are instant and the demo survives a bad network. Anything else goes out to live search and is filed into the same record.

# Ordered, costed, attributed, cited.

HOME LOAN · HDFC BANK  
RESIDENT INDIAN

/checklist – home loan, HDFC Bank

AskExamplesArchitectureHow it works

← Ask a different question

Home loan

HDFC Bank

An HDFC Bank home loan needs three separate bundles: KYC (identity and address), income proof, and property title documents. Property documents are the ones that cause delays — the bank wants an unbroken chain of title, not just the latest deed. HDFC benchmarks its rate to the Policy Repo Rate, so your actual rate sits somewhere inside a wide published band and depends on your credit score, not on what is advertised.

✓ Cached answer

Researched 27 September 2026 · 3 sources read · pre-warmed entry

DOCUMENTS READY

0%0 of 140 of 14 collected

₹ INTEREST RATE BAND

7.75% - 13.20% p.a. (Policy Repo Rate + 2.50% to 7.95%)

Your rate depends on credit score, amount and tenure.

PROCESSING FEE

Up to 0.50% of the loan amount or ₹4,000, whichever is higher, plus applicable taxes, for salaried and self-employed professionals. Up to 1.50% or ₹5,000 for self-employed non-professionals. Non-refundable.

Who regulates what.

The RBI governs the KYC, income-assessment norms and the benchmark

Band, never a single rate

7.75% – 13.20% p.a. Pricing depends on credit score, amount and tenure, so one number would be a lie.

Who regulates what

Every line is labelled **RBI, state stamp act, registrar or bank policy** — and the label is not the model’s opinion.

14 items, each one openable

Where to get it, what it costs, how many days, and which earlier step it depends on.

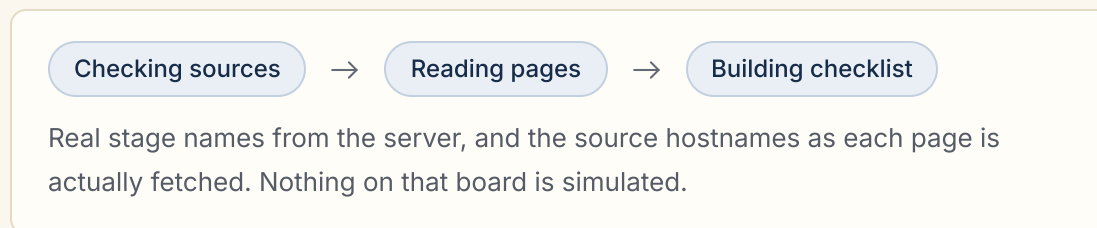
# Cache first. Research on a miss. File the evidence.



## WHY SEARCH AND GENERATION ARE SEPARATE

- ✓ They are **independent failures** — a search outage does not disable structuring.
- ✓ Retrieval grounds the model in **real page text**, not another model's summary.
- ✓ We choose which sources get cited, rather than hoping the model does.

## WHAT THE USER SEES WHILE IT RUNS



**A 202 from DuckDuckGo is a block, not an empty result.** Detected explicitly, given one short retry, then reported as a throttle — never rendered as “no results”.

### FRESHNESS

Document lists move slowly, rates move weekly — so an entry takes the **shorter** window. Every row stores its own TTL.

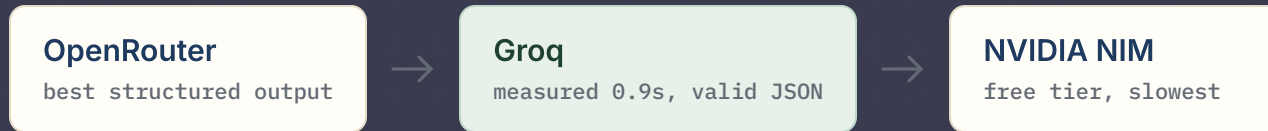
### DEGRADATION ORDER

Fresh hit → stale answer labelled with its age → live research → related cached cases. An error is the last resort, never the first.

### COLD START

On a blank database the schema is created and the seed set filed automatically. There is no setup step to forget.

# One registry, three providers, no single point of failure.



- ✓ **No key is skipped, not called and failed.** An unauthenticated round trip on every request is pure waste.
- ✓ **Out of credit trips the provider out** for five minutes, so no call opens with a guaranteed 402.
- ✓ **Only terminal statuses trip it.** A 500 is one bad call, not grounds for abandoning a provider.
- ✓ **Groq's model list is discovered, not hardcoded** — verified against the live catalogue.
- ✓ **Moderation models are excluded.** A guard model answering a JSON request looks like success and returns nonsense.
- ✓ **Nothing is invented on the way down the chain** — if every provider fails, the app degrades instead of guessing.

GET /api/research/health

```
{ "ok": true,  
  "providers": [ { "name": "openrouter", "has_key": true, "tripped": true },  
                 { "name": "groq", "has_key": true, "tripped": false } ],  
  "has_any_ai_key": true }
```

# It says what it could not confirm.

the app's own disclosure block

## What we could not confirm

- Stamp duty and registration charges are levied by the state government and vary by state and property value; HDFC charges these at actual cost.
- HDFC publishes a minimum rate rather than a customer-specific rate. The rate you are offered depends on your credit score and profile.

sources read for this answer

## Sources read for this answer

3 of them

- HDFC Bank — Home Loan Required Documents, Charges, Processing Fee 2026  
homeloans.hdfc.bank.in
- HDFC Bank — Home Loan product page and rate table  
homeloans.hdfc.bank.in
- HDFC Bank — Home Loan Interest Rate  
homeloans.hdfc.bank.in

Both panels are lifted straight from one result page, unmodified.

### Never invented

A cost or rate with no source becomes **“not publicly disclosed — contact the bank directly.”** Inverted and absurd ranges are repaired, not prettied.

### Never misattributed

Stamp duty and registration are re-filed to the state or registrar **after** generation. The prompt asks for the right answer; the code guarantees it.

### Bands, not guarantees

Real pricing depends on credit score, amount and tenure, so every rate is a published band.

# Any answer can be taken apart.

THE CACHE ROW IS THE AUDIT TRAIL

- ✓ The **exact query** that produced this answer.
- ✓ Every **source URL** that was actually read.
- ✓ **When** it was researched, and when it goes stale.
- ✓ Which **model and provider** structured it.

Nothing is cited that was not opened. A URL only becomes a source if the fetch returned readable text — so the citation list is exactly the set of pages that were read.

WHY THE CACHE IS TRUSTWORTHY

A pre-warmed entry is marked as such on screen and is not presented as a live lookup. The freshness window is stored per row, so a rate cannot outlive the page it came from.

● ● ● show the exact research record

☰ [Show the exact research record](#)

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Researched          | 2026-09-27T11:03:16.278404+00:00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Cache key           | home loan hdfc bank resident                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Query sent          | What documents, in what order, with what approximate costs and how many working days each step takes, are required for Home loan at HDFC Bank for a resident Indian applicant (ordinary resident, taxed in India)? Also state the current interest rate range as a band (for example "8.50% - 9.75% p.a.") and the processing fee. Note separately which items come from RBI rules, which from the state stamp act or registrar, and which are the bank's own policy. If a detail is not publicly disclosed, say so explicitly rather than estimating. Cite every source. |
| Search model        | seed-research                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Re-researched after | 30 days                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

# Tested like infrastructure, not like a demo.

188

Python tests, no network and no API keys required

21/21

page × width combinations clean

14/14

example links return sourced checklists

12/12

keyboard and form-submission checks

## THE GATES EARNED THEIR KEEP

- ✓ **A boot crash.** DATABASE\_PATH in a missing directory died on the first write.
- ✓ **A form that went nowhere** — the submit button reloaded the home page, so it looked like “no results”.
- ✓ **A contrast failure** at 4.41:1, and a board clipping off-canvas at 375px.

## WHAT RUNS IN CI

- ✓ Full Python suite, then a **real browser boot**, before anything is deployed.
- ✓ Flip-board geometry asserted across **0–14 documents** and all three statuses.
- ✓ No dependency on a public search engine having a good day.

pytest tests/ -q

188 passed in ~7s

npm run gates

21/21 · stable · 14/14 · 12/12

# An engineering partner, start to finish.

## DESIGNED WITH BOB

- ✓ The **cache-first architecture** — separating search from generation so both can fail independently.
- ✓ The **provider registry**, circuit breaker, and runtime model discovery.
- ✓ **SQLite caching, job polling**, search-result parsing, page ranking.

## DIAGNOSED WITH BOB

- ✓ The form that reloaded instead of searching, and an **application-context error** in a background thread.
- ✓ A **residency-code mismatch** that silently emptied failure suggestions.
- ✓ **A regression test per defect** — the boot crash, the form that reloaded instead of searching, and the empty-suggestion mismatch each have a test named after them.

## WHERE TO CHECK IT IN THE REPOSITORY

`app/ai/client.py`

The provider registry, circuit breaker and model discovery.

`app/research/agent.py`

Cache-first orchestration and the degradation ladder.

`app/research/synthesize.py`

The honesty rules enforced after generation.

`tests/`

188 tests plus four browser gates, one per defect class.

No watsonx.ai

No watsonx Orchestrate

Task-session summaries in the repository

## HONEST LIMITS

# What this is, and is not.

### KNOWN LIMITS

- ✓ Informational only. **Not legal or financial advice.**
- ✓ Cold research needs a reachable search provider; when blocked it **shows overlapping cached cases** instead of failing blank.
- ✓ English only. Coverage is exactly as good as **public sources** are.

### NEXT

- ✓ Persist the research record so the cache survives restarts.
- ✓ **Per-field freshness** — rates move weekly, document lists monthly.
- ✓ Regional languages, and upload-based verification.

### TRADEOFFS WE CHOSE DELIBERATELY

#### SQLite, not Postgres

One writer, one process, one file. Correct for this. Scaling out means migrating off it — a different project, not a config change.

#### Keyless search, not a paid API

Costs nothing and works with no keys, which is why the whole app runs free. The trade is that it can be rate-limited.

#### Grounding, not a bigger model

Reading the real pages beats asking a larger model to summarise them, and it is what makes the citations checkable.

## Ask it anything. It will show its work.

Every figure traceable to a source you can open, and every gap named instead of filled.

[github.com/gabsgj/Kaagaz](https://github.com/gabsgj/Kaagaz)

flask · sqlite · no build step